2. For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in python to output a description of the set of all hypotheses consistent with the training examples.

CODING:

```
import pandas as pd

data = pd.DataFrame([
  ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
  ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
  ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
  ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
], columns=['Sky', 'Temp', 'Humidity', 'Wind',
       'Water', 'Forecast', 'EnjoySport'])

S = ['0'] * 6
G = [['?'] * 6]

for _, row in data.iterrows():
  x = row.iloc[:-1].tolist()

  if row.iloc[-1] == 'Yes':
    for i in range(6):
      if S[i] == '0':
        S[i] = x[i]
      elif S[i] != x[i]:
        S[i] = '?'

    G = [g for g in G if all(
```

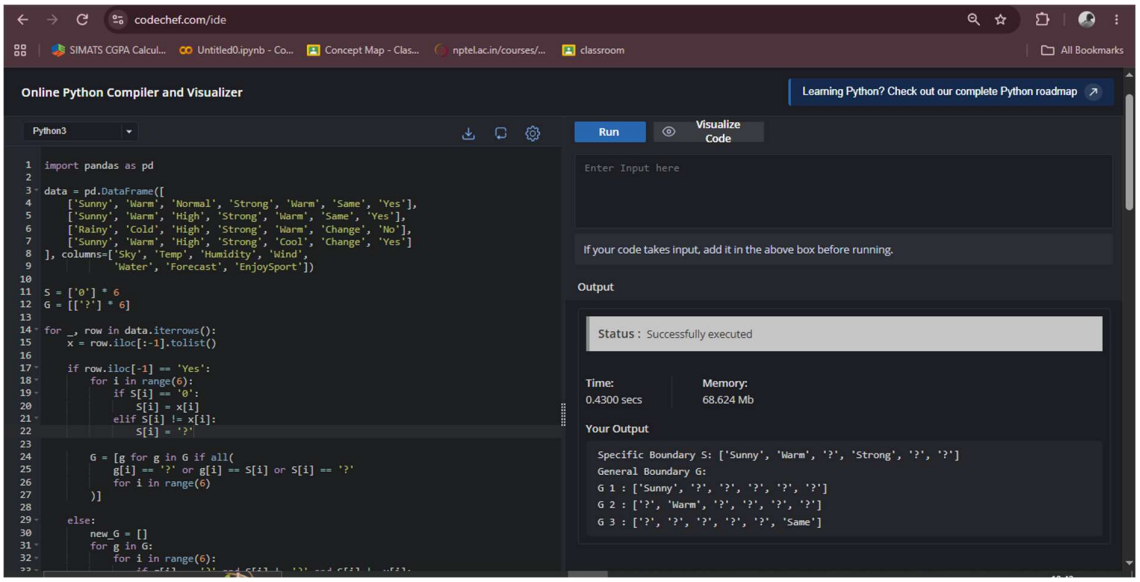
```
            g[i] == '?' or g[i] == S[i] or S[i] == '?'
            for i in range(6)
        )]

    else:
        new_G = []
        for g in G:
            for i in range(6):
                if g[i] == '?' and S[i] != '?' and S[i] != x[i]:
                    h = g.copy()
                    h[i] = S[i]
                    new_G.append(h)
        G = new_G if new_G else G

print("Specific Boundary S:", S)

print("General Boundary G:")
for i, g in enumerate(G, 1):
    print("G", i, ":", g)
```

OUTPUT:


codechef.com/ide
SIMATS CGPA Calcul...
Untitled0.ipynb - Co...
Concept Map - Clas...
nptel.ac.in/courses/...
classroom
All Bookmarks
Online Python Compiler and Visualizer
Learning Python? Check out our complete Python roadmap
Python3
Run
Visualize Code
Enter Input here
If your code takes input, add it in the above box before running.
Output
Status : Successfully executed
Time:
0.4300 secs
Memory:
68.624 Mb
Your Output
Specific Boundary S: ['Sunny', 'Warm', '?', 'Strong', '?', '?']
General Boundary G:
G 1 : ['Sunny', '?', '?', '?', '?', '?']
G 2 : ['?', 'Warm', '?', '?', '?', '?']
G 3 : ['?', '?', '?', '?', '?', 'Same']